



SUBDIVISIONS OF BIOLOGY — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Zoology — Scope & Father of Zoology
2. Core Branches — Ornithology, Genetics, Ecology & Palaeobiology
3. Specialized Branches — Flowers, Skull, Grasses, Pollen & Invertebrates
4. Population, Culture & Behaviour — Demography, Ethnology & Ethology
5. Bionics & Related Terms — Bionomics, Bionomy & Biometry
6. Palaeobotany — Plant Fossils & Birbal Sahni
7. Animal Rearing & Production — Sericulture, Apiculture, Pisciculture & Lac Culture
8. Plant Cultivation Branches — Horticulture, Olericulture, Pomology, Floriculture & Agronomy
9. Lexicography & Related Terms
10. Viticulture — Grape Cultivation
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13. Gerontology — Old Age & Ageing
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15. Snake Studies & Other Zoology Branches
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1. Zoology — Scope & Father of Zoology

- Zoology biology ki branch hai jo living aur extinct — dono animals ka study karti hai; isme structure, embryology, evolution, classification, habits aur distribution include hote hain. [Q1]
- Aristotle ko source 'Father of Zoology' batata hai. [Q1]

2. Core Branches — Ornithology, Genetics, Ecology & Palaeobiology

- Birds ka study — Ornithology; Inheritance ka study — Genetics; Environment / organisms-environment relationship ka study — Ecology; Fossils ka study — Palaeobiology. [Q2]
- Match-the-following order source mein: Birds → Ornithology (A-3), Inheritance → Genetics (B-4), Environment → Ecology (C-2), Fossil → Palaeobiology (D-1). [Q2]

3. Specialized Branches — Flowers, Skull, Grasses, Pollen & Invertebrates

- Source ke mutabik flowers ka study — Anthology; human skull ka study — Phrenology; grasses ka study — Agrostology; pollen grains aur other spores ka study — Palynology. [Q3]
- Insects ka scientific study — Entomology; fish ka study — Ichthyology; parasitic organisms ka study — Parasitology; molluscs aur unke shells ka study — Malacology. [Q4]

4. Population, Culture & Behaviour — Demography, Ethnology & Ethology

- Demography — populations, including human beings, ka statistical study. Ecology — organisms aur environment ke interrelationship ka study; Genetics — heredity and variation; Virology — viruses ka study. [Q5]
- Ethnology — different races/cultures of mankind ka comparative study; Ethnography — ek specific culture ka study; Ethology — animal behaviour; Ethics — ethical duties aur moral principles ka study. [Q6]

5. Bionics & Related Terms — Bionomics, Bionomy & Biometry

- Bionics — nature mein biological systems ke properties/methods ka study aur us knowledge ka mechanical world / engineering mein use. [Q7]
- Bionomics — organism aur environment ke relation ka comprehensive study; Bionomy — laws of life ka study; Biometry — biological problems ka statistical study. [Q7]

6. Palaeobotany — Plant Fossils & Birbal Sahni

- Palaeobotany palaeontology ki branch hai jo geological contexts se plant fossils — ancient plants ke remains — ki recovery aur identification se related hai. [Q8]
- Q8 ke options mein plant fossils explicitly nahi the, isliye source correct choice (D) 'None of these' deta hai. Source Prof. Birbal Sahni ko Father of Indian Palaeobotany batata hai. [Q8]

7. Animal Rearing & Production — Sericulture, Apiculture, Pisciculture & Lac Culture

- Sericulture — silk production ke liye silkworms ka rearing. Apiculture — honey/wax production ke liye honey bees ka rearing. [Q9, Q10]
- Pisciculture fish farming / fish hatchery se related hai; lac insect ke rearing ko Lac culture kaha jata hai. [Q13]
- Q13 mein 'Pisciculture — Lac insect' pair source ke hisab se incorrectly matched hai. [Q13]

8. Plant Cultivation Branches — Horticulture, Olericulture, Pomology, Floriculture & Agronomy

- Horticulture ek broader agriculture branch hai jisme Pomology (fruits), Olericulture (vegetables) aur Floriculture (flowers) include hote hain. [Q11]
- Olericulture specifically vegetable growing ka science hai — non-woody/herbaceous plants ko food ke liye cultivate karna. [Q12]
- Floriculture — flowers; Pomology — fruits; Agronomy — crops. [Q11]

⚠ EXAM TRAP — Horticulture vs Olericulture — Q11–Q12

- ▶ Q11 ke specific matching context mein vegetables ke liye 'Horticulture' ko incorrect maana gaya, kyunki exact term Olericulture hai. [Q11]
- ▶ General sense mein Horticulture broader field hai; source explanation itself Olericulture ko vegetables ka specific branch batata hai. [Q11, Q12]

9. Lexicography & Related Terms

- Lexicography — dictionaries ki design, compilation aur evaluation; source pair 'Lexicography — Compilation of dictionaries' ko correctly matched batata hai. [Q14, Q15]
- Philology — language structure aur history; stamp collection — Philately; Entomology — insects; Iconography — idols / visual arts ka study. [Q14]

10. Viticulture — Grape Cultivation

- Viticulture grapes ki science, production aur study hai; vineyard mein hone wale processes/events se related hai. [Q16]

11. Vermiculture — Earthworm Culture

- Vermiculture — earthworms ko manage aur cultivate / raise karne ka process. [Q17]
- Earthworms organic waste ko nutrient-rich soil / vermicompost mein convert karne mein help karte hain; vermiculture mein earthworms hi use hote hain. [Q17, Q18]

12. Genetics — Heredity & Variation

- Genetics genes, heredity aur living organisms mein genetic variations ka broad study hai. [Q19]
- Mendel's laws aur DNA structure genetics ke parts hain, lekin source ke mutabik most comprehensive definition heredity and variations ka study hai. [Q19]

13. Gerontology — Old Age & Ageing

- Gerontology — old age, ageing process aur old people ke particular problems ka scientific study; isme social, psychological, cognitive aur biological aspects of ageing include hote hain. [Q20, Q21, Q22]
- Oncology — cancer treatment se related medical branch; Teratology — physiological development ki abnormalities ka study; Ornithology — birds ka study. [Q21]
- Thanatology — death aur usse associated practices ka scientific study. [Q22]

14. Pedology & Osteology — Soil and Bones

- Pedology — natural environment mein soil formation, soil morphology aur soil classification ka study. [Q23]
- Osteology — bones ka study. Orology — mountains; Serology — serum aur other body fluids; Geology — earth ki internal structure aur andar milne wale matter ka study. [Q24, Q25]

15. Snake Studies & Other Zoology Branches

- Serpentology / Ophiology — snakes ka study; source ise Zoology ki branch batata hai. [Q26]
- Herpetology broader branch hai jo amphibians aur reptiles — frogs, toads, salamanders, snakes, lizards, turtles, crocodiles — se related hai; Serpentology iski sub-discipline hai. [Q26]
- Ornithology — birds; Ichthyology — fish. [Q26]

16. Core Biology vs Psychology + Hydroponics

- Physiology, Pathology aur Bacteriology source ke according biology/medical science ke core branches hain; Psychology mind aur behaviour ka study hai, isliye Q27 mein odd one out hai. [Q27]
- Hydroponics hydroculture ka subset hai — mineral nutrient solutions in water ka use karke plants ko soil ke bina grow karna. [Q28]